Higher National Diploma in Engineering (Mechanical/Civil)

Higher National Diploma in Building Service Engineering

First Year, First Semester Examination – 2020

CE-1102-Fluid Mechanics

Answer any five questions. **3hours**

Question 1 (Q1)

a) Explain the terms "**Total pressure**" and "**Centre of pressure**" related to a plate submerged in a static fluid and write equations to find the Total pressure and center of pressure of an inclined plate submerged in a static fluid. Support your answer with clear sketches. (06 marks)

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b) A **2 m wide gate** supporting water is shown in Fig. 01. Find the height **h** of the water so that the gate tips about the hinge. (Hint: - Gate tips if resultant pressure force passes through a point in the range BC of the gate) . (14 marks)

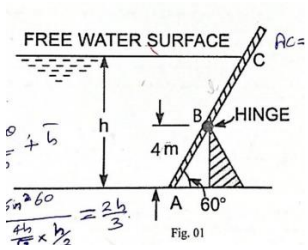

Fig. 01

Question 2 (Q2)

a) Explain what does it mean by "**viscosity**" in fluid mechanics. (02 marks)

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b) What does it mean by "**Ideal Fluid**" & "**Real Fluid**"? (02 marks)

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c) Two large plane surfaces are **2 cm** apart and the space between the surfaces is filled with oil (**Fig. 02**). What force (F) is required to drag a very thin plate of surface area $\mathbf{0.5\ m^2}$ between the two large plane surfaces at a speed of $\mathbf{0.5\ m/s}$, if the thin plate is in the middle of the two plane surfaces? Take dynamic viscosity of oil $= \mathbf{0.8\ N \cdot s/m^2}$. (12 marks)

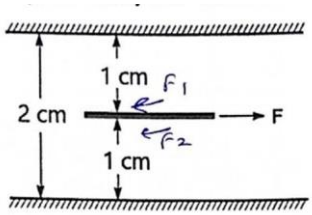

Fig. 02

d) Explain the operating principle of an **orifice type viscosity meter** with suitable sketches. (04 marks)

Question 3 (Q3)

a) Explain the **continuity equation**. (02 marks)

b) Explain **Bernoulli's equation**. (02 marks)

c) A horizontal venturi meter with inlet and throat diameters **32 cm** and **16 cm** respectively is used to measure the flow of water. The reading of the differential manometer connected to the inlet and throat is **18 cm of mercury**. Determined the rate of flow. take $\mathbf{C_d = 0.98}$. (10 marks)

d) Describe the operating principle of the **Pitot tube** with suitable sketches and relevant equations. (06 marks)

Question 4 (Q4)

a) What is the **major cause of energy loss** of fluid flow in pipes? (01 marks)

b) Write **four minor causes of energy loss** of fluid flow in pipes. (03 marks)

c) The rate of flow of water through a horizontal pipe is $\mathbf{0.24\ m^3/s}$. The diameter of the pipe is **180 mm** is suddenly enlarged to **300 mm**. The pressure intensity in the smaller pipe is $\mathbf{11.5\ N/cm^2}$. Determine: i) loss of head due to sudden enlargement ii) pressure intensity in the large pipe iii) power lost due to enlargement (11 marks)

d) Three pipes of lengths $\mathbf{750\ m}$, $\mathbf{500\ m}$ and $\mathbf{420\ m}$ and diameters $\mathbf{500\ mm}$, $\mathbf{450\ mm}$ and $\mathbf{350\ mm}$ respectively are connected in **series**. These pipes are to be replaced by a **single pipe of length $\mathbf{1600\ m}$**. Find the diameter of the single pipe. (04 marks)

Question 5 (Q5)

a) What does it mean by **drag force**? (02 marks)

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b) What are the **two types of drag forces**? (02 marks)

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c) With clear sketches explain a **streamlined body** and a **bluff body**. (04 marks)

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d) A partially submerged body is towed in water. The resistance R to its motion depends on the density ρ, viscosity μ of the water, length L of the body, velocity V of the body and the acceleration due to gravity g. Show that the resistance to the motion can be expressed in the form:

$$R = \rho\, L^2 V^2 \phi \left[\left(\frac{\mu}{\rho V L} \right) \cdot \left(\frac{Lg}{V^2} \right) \right]$$

(12 marks)

Question 6 (Q6)

a) A jet of water of diameter **80 mm** moving with a velocity of $\mathbf{30\ m/s}$ strikes a **fixed plate** in such a way that the angle between the jet and the plate is $\mathbf{60}^\circ$. Find the force exerted by the jet on the plate: i) in the direction normal to the plate ii) in the direction of the jet (08 marks)

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b) Draw a clear sketch and name the main parts of a **centrifugal pump**. (04 marks)

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c) Draw a clear sketch and name the main parts of a **single-acting reciprocating pump**. (04 marks)

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d) Draw a clear sketch and name the main parts of a **Pelton turbine**. (04 marks)